

- 4 (a) State what is meant by resonance.

.....

.....

..... [2]

- (b) A small ball is held in place using a stretched string. One end of the string is fixed to a wall and the other end is attached to a vibration generator, as shown in Fig. 4.1.

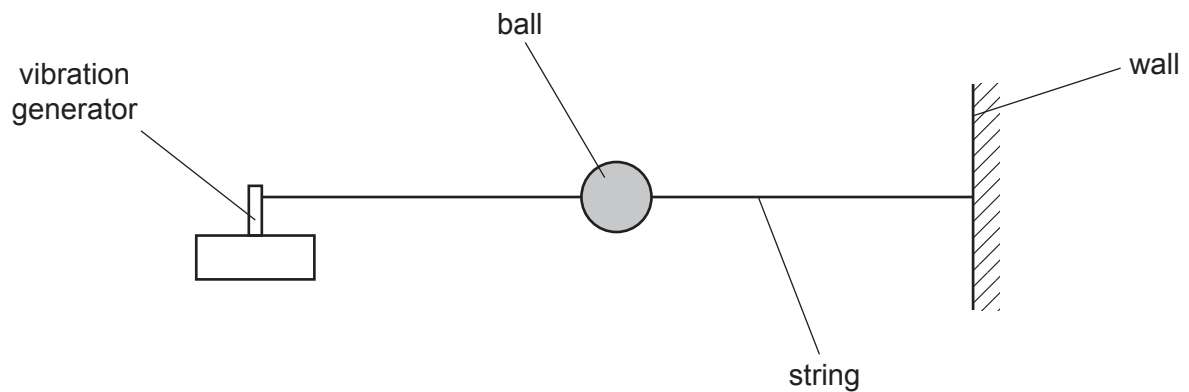


Fig. 4.1

Initially, the vibration generator is switched off.

A student displaces the ball vertically and then releases it. Fig. 4.2 shows the variation of the displacement of the ball with time after it is released.

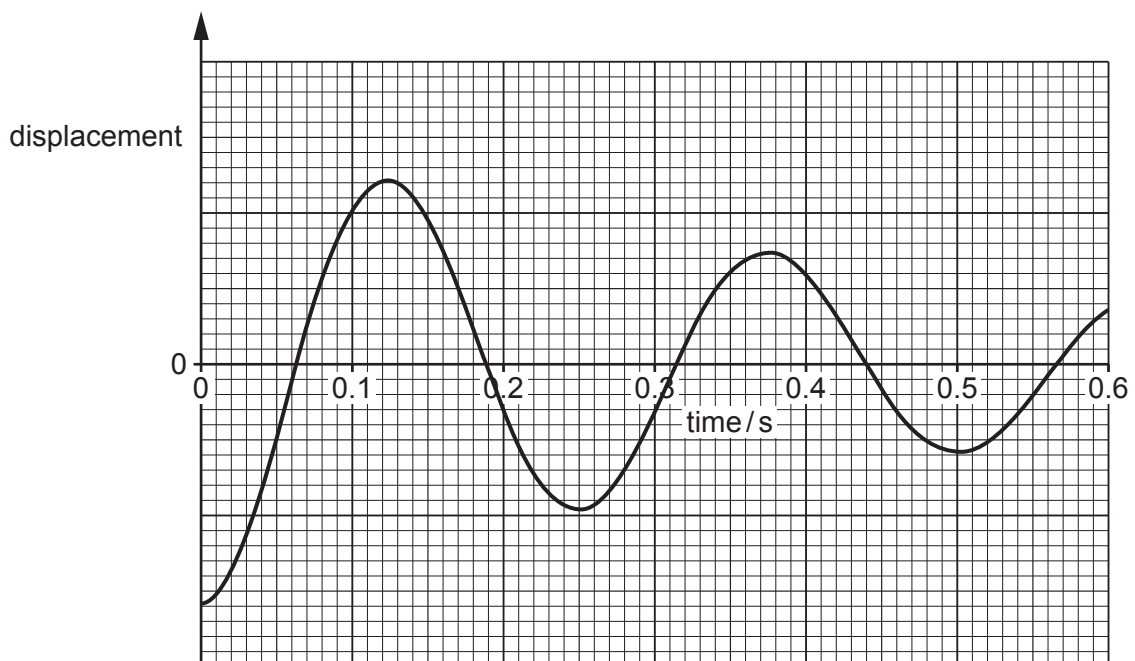


Fig. 4.2

- (i) State the name of the phenomenon illustrated by the decrease in the amplitude of the oscillations in Fig. 4.2.

..... [1]

- (ii) Explain the decrease with time of the amplitude of the oscillations of the ball.

.....

 [2]

- (iii) Determine the frequency of the oscillations of the ball.

frequency = Hz [1]

- (c) The vibration generator in (b) is switched on and its frequency f of vibration is gradually increased from 0 to 10 Hz.

On Fig. 4.3, sketch the variation with f of the amplitude of the oscillations of the ball.

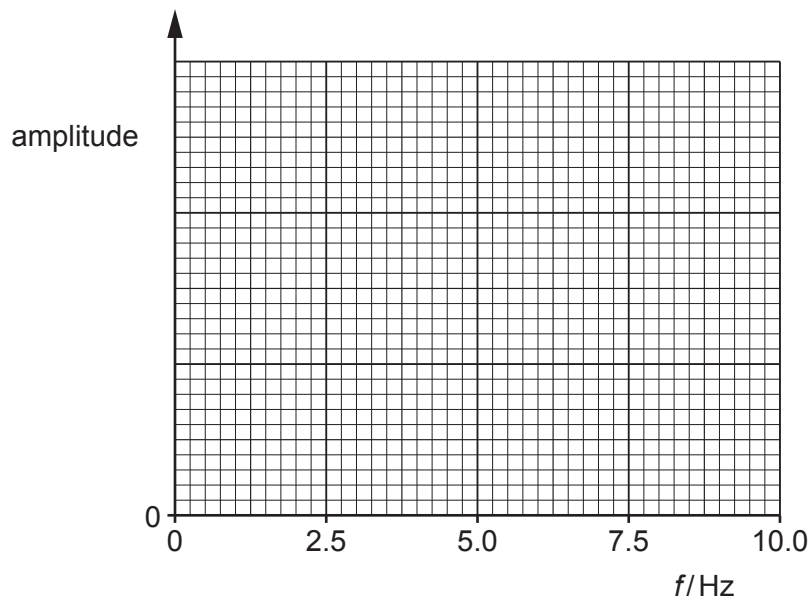


Fig. 4.3

[2]

[Total: 8]